

```
import random
```

```
def line(symbol, size):  
    print(symbol * size)
```

```
def swap(a, b):  
    return b, a
```

```
def play_innings(batting_team, bowling_team, total_overs, target=None):  
    batsman_runs = [0] * 11  
    batsman_balls = [0] * 11  
    bowler_runs = [0] * 11  
    bowler_balls = [0] * 11  
    bowler_wickets = [0] * 11  
  
    wicket = 0  
    live_score = 0  
    striker = 0  
    non_striker = 1  
    next_player = 2  
    current_over = 0  
    current_ball = 0  
    bowler_index = 10
```

```

print("Enter runs (0-6) or 'W' for wicket.\n")

while wicket < 10 and current_over < total_overs:

    if target and live_score >= target:

        break

    if target:

        balls_remaining = (total_overs * 6) - (current_over * 6 + current_ball)

        print(f"\nOver {current_over}.{current_ball} | Score: {live_score}/{wicket} | "

              f"Need: {target - live_score} off {balls_remaining} | "

              f"Batsmen: {batting_team[striker]} "

              f"{batsman_runs[striker]}({batsman_balls[striker]}) "

              f"& {batting_team[non_striker]} "

              f"{batsman_runs[non_striker]}({batsman_balls[non_striker]}) | "

              f"Bowler: {bowling_team[bowler_index]}")

    else:

        print(f"\nOver {current_over}.{current_ball} | Score: {live_score}/{wicket} | "

              f"Batsmen: {batting_team[striker]} "

              f"{batsman_runs[striker]}({batsman_balls[striker]}) "

              f"& {batting_team[non_striker]} "

              f"{batsman_runs[non_striker]}({batsman_balls[non_striker]}) | "

              f"Bowler: {bowling_team[bowler_index]}")

    user_input = input(">> ").strip().upper()

    if user_input == 'W':

```

```
print(f"{batting_team[striker]} OUT!")

wicket += 1

batsman_balls[striker] += 1

bowler_wickets[bowler_index] += 1

bowler_balls[bowler_index] += 1

current_ball += 1

if next_player < 11:

    striker = next_player

    next_player += 1

else:

    break


elif user_input in '0123456':

    run = int(user_input)

    live_score += run

    batsman_runs[striker] += run

    bowler_runs[bowler_index] += run

    batsman_balls[striker] += 1

    bowler_balls[bowler_index] += 1

    current_ball += 1

    if run % 2 == 1:

        striker, non_striker = swap(striker, non_striker)

else:

    print("Invalid input! Enter 0-6 or W.")

    continue
```

```
if current_ball == 6:
    current_over += 1
    current_ball = 0
    striker, non_striker = swap(striker, non_striker)
    bowler_index = (bowler_index - 1 + 11) % 11
    print(f"---- End of Over {current_over} ----")
```

```
return {
    "score": live_score,
    "wickets": wicket,
    "overs": current_over,
    "balls": current_ball,
    "batsman_runs": batsman_runs,
    "batsman_balls": batsman_balls,
    "bowler_runs": bowler_runs,
    "bowler_balls": bowler_balls,
    "bowler_wickets": bowler_wickets,
}
```

```
def print_scorecard(batting_team, bowling_team, innings_data):
```

```
    print(f"\n{'Batting:'}")
```

```
    print(f"{'Batsman':<40}{'Runs':<15}{'Balls':<15}")
```

```
    for i in range(11):
```

```
        print(f"{batting_team[i]:<40}{innings_data['batsman_runs'][i]:<15}{innings_data['batsman_balls'][i]:<15}")
```

```

print(f"\n{'Bowling:'}")

print(f"{'Bowler':<40}{'Overs':<15}{'Runs':<15}{'Wickets':<15}")

for i in range(11):

    if innings_data['bowler_balls'][i] > 0:

        overs_str = f"{innings_data['bowler_balls'][i] // 6}.{innings_data['bowler_balls'][i] % 6}"

print(f"{'bowling_team[i]:<40}{overs_str:<15}{innings_data['bowler_runs'][i]:<15}{innings_data['bowler_wickets'][i]:<15}")


def main():

    # ---- Enter Team Names and Players ----

    team_one_name = input("Enter the name of the first team: ").strip()

    print("Enter the names of players:")

    team1 = []

    for i in range(11):

        player = input(f"{i+1}.\t").strip()

        team1.append(player)


    team_two_name = input("Enter the name of the second team: ").strip()

    print("Enter the names of players:")

    team2 = []

    for i in range(11):

        player = input(f"{i+1}.\t").strip()

        team2.append(player)

```

```

# ---- Display Players Side by Side ----

line('-', 100)

print(f"{'':}{team_one_name:<50}{team_two_name:<50}")

line('-', 100)

print()

for i in range(11):

    print(f"{team1[i]:<50}{team2[i]:<50}")


# ---- Toss ----

toss = random.randint(0, 1) # 0 = Heads, 1 = Tails


team1_call = input(f"\n{team_one_name}: Heads or Tails?\nPress 'H' for Heads and 'T' for
Tails: ").strip().upper()


if team1_call not in ['H', 'T']:

    print("ERROR! Invalid input.")

    return


team2_call = 'T' if team1_call == 'H' else 'H'

print(f"{team_one_name} has called {'Heads' if team1_call == 'H' else 'Tails'}.")
print(f"{team_two_name} has called {'Heads' if team2_call == 'H' else 'Tails'}.")
print(f"Toss result: {'Heads' if toss == 0 else 'Tails'}")


team_one_won_toss = (toss == 0 and team1_call == 'H') or (toss == 1 and team1_call ==
'T')


if team_one_won_toss:

```

```
print(f"{team_one_name} has won the toss!\n{team_one_name} will bat first.")

batting_team = team1

bowling_team = team2

batting_team_name = team_one_name

bowling_team_name = team_two_name

else:

    print(f"{team_two_name} has won the toss!\n{team_two_name} will bat first.")

    batting_team = team2

    bowling_team = team1

    batting_team_name = team_two_name

    bowling_team_name = team_one_name


# ---- Number of Overs ----

total_overs = int(input("Enter number of overs: "))


# ---- First Innings ----

print()

line('-', 43)

print("FIRST INNINGS")

line('-', 43)

print()

first_innings = play_innings(batting_team, bowling_team, total_overs)

target = first_innings["score"] + 1


line('-', 39)
```

```
print("FIRST INNINGS SCORECARD")
```

```
line('-', 39)
```

```
print_scorecard(batting_team, bowling_team, first_innings)
```

```
print(f"\nTARGET: {target}\n")
```

```
# ---- Second Innings ----
```

```
line('-', 43)
```

```
print("SECOND INNINGS")
```

```
line('-', 43)
```

```
print()
```

```
second_innings = play_innings(bowling_team, batting_team, total_overs, target)
```

```
print()
```

```
line('-', 38)
```

```
print("SECOND INNINGS SCORECARD")
```

```
line('-', 38)
```

```
print_scorecard(bowling_team, batting_team, second_innings)
```

```
# ---- Match Summary ----
```

```
print()
```

```
line('-', 43)
```

```
print("MATCH SUMMARY")
```

```
line('-', 43)
```

```
print()
```



```

chasing_team_name = bowling_team_name
defending_team_name = batting_team_name

print(f"\n{'TEAM':<20}{'SCORE':<15}{'WICKETS':<10}{'OVERS':<10}")
line('-', 60)

print(f"\n{batting_team_name:<20}{first_innings['score']:<15}{first_innings['wickets']:<10}"
      f"{first_innings['overs']}.{first_innings['balls']}")

print(f"{chasing_team_name:<20}{second_innings['score']:<15}{second_innings['wickets']:<10}"
      f"{second_innings['overs']}.{second_innings['balls']}")
line('-', 60)
print()

score2 = second_innings["score"]
wicket2 = second_innings["wickets"]

if score2 >= target:
    print(f"{chasing_team_name} won by {10 - wicket2} wickets.")
elif score2 == target - 1:
    print("Match tied.")
else:
    print(f"{defending_team_name} won by {target - 1 - score2} runs.")

# ---- Man of the Match ----
best_runs = -1

```

```
mom = ""
```

```
if score2 >= target:
```

```
    for i in range(11):
```

```
        if second_innings["batsman_runs"][i] > best_runs:
```

```
            best_runs = second_innings["batsman_runs"][i]
```

```
            mom = bowling_team[i]
```

```
else:
```

```
    for i in range(11):
```

```
        if first_innings["batsman_runs"][i] > best_runs:
```

```
            best_runs = first_innings["batsman_runs"][i]
```

```
            mom = batting_team[i]
```

```
print(f"\nMAN OF THE MATCH: {mom} ({best_runs} runs)")
```

```
if __name__ == "__main__":
```

```
    main()
```